Appointment API

Overview

The Appointment resource provides the ability to book & cancel appointments and retrieve information (tele-visit and e-Check In links) about appointments.

Implementation Notes:
When integrating your applications, please work with the client to determine the providers and facility IDs which they want to make available to your applications for enabling scheduling functionality.

Authorization Types:
System

Booking appointments

HTTP method:
POST

URL:
v1 = https://{FQDN}/apps/api/v1/fhir/{PracticeCode}/dstu2/Appointment
v2 = https://{FQDN}/apps/api/v1/fhir/{PracticeCode}/dstu2/v2/Appointment

Headers:
Authorization: <OAuth2 Bearer Token>
Content-Type: application/fhir+json

Request Body:

Name	Required?	Type	Description
patient resource	yes	object	Patient for whom appointment is being booked. (First Name, Last Name, DOB, Phone & Email are used for patient matching)
slot	yes	string	Slot id from slot API
reason	no	string	Reason text for the appointment
description	no	string	Description text for the appointment
comments	no	string	Comment text for the appointment

Sample:

Request Headers:
Accept: application/json+fhir
Authorization: <OAuth2 Bearer Token>

POST http://healow-preprod.healow.com/apps/api/v1/fhir/IFDECD/fhir/Appointment

Body:


```
{
  "resourceType": "Appointment",
  "contained": [
    {
      "resourceType": "Patient",
      "id": "patA",
      "name": [{
        "use": "usual",
        "family": ["testfamily"],
        "given": ["testgiven"],
        "suffix": ["MSc"]
```

```
}],
"telecom": [
    {
        "system": "phone",
        "value": "0000000000",
        "use": "mobile"
    },
    {
        "system": "email",
        "value": "test@example.com",
        "use": "home"
    }
],
"gender": "male",
"birthDate": "1970-01-01",
"maritalStatus": {
    "coding": [{
        "system": "http://hl7.org/fhir/v3/MaritalStatus",
        "code": "M",
        "display": "Married"
    }],
    "text": "Getrouwd"
},
"multipleBirthBoolean": true,
"contact": [{
    "relationship": [{"coding": [{
        "system": "http://hl7.org/fhir/patient-contact-relationship",
        "code": "partner"
    }]}],
    "name": {
        "use": "usual",
        "family": ["testfamilycontact"],
        "given": ["testgivencontact"]
    },
    "telecom": [{
        "system": "phone",
        "value": "0000000000",
        "use": "mobile"
    }]
}],
"address": [{
    "use": "home",
    "type": "both",
    "text": "132, My Street, Kingston, NY, 12401, US",
    "line": [
        "132",
        "My Street"
    ],
    "city": "Kingston",
    "state": "NY",
    "postalCode": "12401",
    "country": "US"
}],
"communication": [{
    "language": {
        "coding": [{
            "system": "urn:ietf:bcp:47",
            "code": "nl",
            "_code": {"fhir_comments": ["    IETF language tag    "]},
            "display": "Dutch"
        }],
        "text": "Nederlands"
    },
    "preferred": true
}],
"extension": [
    {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
        "extension": [
            {
```

```
                "url": "ombCategory",
                "valueCoding": {
                    "system": "urn:oid:2.16.840.1.113883.6.238",
                    "code": "1002-5",
                    "display": "American Indian or Alaska Native"
                }
            },
            {
                "url": "detailed",
                "valueCoding": {
                    "system": "urn:oid:2.16.840.1.113883.6.238",
                    "code": "1442-3",
                    "display": "Indian Township"
                }
            },
            {
                "url": "text",
                "valueString": "American Indian or Alaska Native"
            }
        ]
    },
    {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
        "extension": [
            {
                "url": "detailed",
                "valueCoding": {
                    "system": "urn:oid:2.16.840.1.113883.6.238",
                    "code": "2155-0",
                    "display": "Central American"
                }
            },
            {
                "url": "ombCategory",
                "valueCoding": {
                    "system": "urn:oid:2.16.840.1.113883.6.238",
                    "code": "2135-2",
                    "display": "Hispanic or Latino"
                }
            },
            {
                "url": "text",
                "valueString": "Hispanic or Latino"
            }
        ]
    }
  ],
  "managingOrganization": {
    "reference": "Organization/f001",
    "display": "Burgers University Medical Centre"
  }
},
{
  "resourceType": "Coverage",
  "type": {
    "coding": [{
      "system": "eCW_insurance_coding_system(Cash/Insurance/Not-Applicable)",
      "code": "insurance_code",
      "display": "insurance name"
    }],
    "text": "insurance name"
  }
}
],
"id": "",
"status": "proposed",
"reason": {"text": "The reason that this appointment is being scheduled. (e.g. Regular checkup)"},
"description": "The brief description of the appointment as would be shown on a subject line in a meeting request, or appointment list.",
"slot": [{"reference": "<provide slot Id here>"}],
"comment": "Additional comments about the appointment.",
```

```
  "participant": [
    {
      "actor": {"reference": "Practitioner/1234564789"},
      "required": "required"
    },
    {
      "actor": {"reference": "#patA"},
      "required": "required"
    }
  ]
}
```


Response:

v1:


```
{
  "resourceType": "OperationOutcome",
  "text": {"div": "Your appointment has been reserved for 2023-03-20 12:00:00. Reference ID is 3bd3a5ee-43be-4cc3-9cf0-ae81cc16b564. Appt Block Id is 0. Appt Encounter Id is 30435."}
}
```


v2:


```
{
  "resourceType": "Bundle",
  "id": "9a93ce07-0cf2d2b8-e289-56909ca9d8a4",
  "type": "transaction-response",
  "entry": [
    {
      "resourceType": "OperationOutcome",
      "text": {"div": "Your appointment has been reserved for 2021-02-25 10:00:00. Reference ID is 9a93ce07-0cf2d2b8-e289-56909ca9d8a4 . Appt Block Id is 0. Appt Encounter Id is 29967 ."}
    },
    {"response": {
      "status": "1",
      "location": "Encounter/889f6f9b-05cc-4bb8-8206-67eb07faaad9"
    }},
    {"response": {
      "status": "1",
      "location": "Patient/NwOe2E9laWqwTWAcStx4RTu3PKsEwwyDsPLnAYJJelc"
    }}
  ]
}
```


Error Handling:

In the event of an error, the system will respond by supporting below error codes.

Supported HTTP codes:

HTTP Code	Severity	Description	Example
401	error	Unauthorized	Invalid Access Token
403	error	Forbidden	Valid Access Token but not authorized for the requested scope
408	error	Request time out	Service timed out
500	error	Internal Server error	Server issue

400	error	Bad request	Incomplete or invalid search parameter combinations.

Sample error response:


```
{"error": {
   "message": "An access token is required to request this resource.",
   "type": "OAuthException",
   "reason": "unauthorized"
}}
```

```
{
   "resourceType": "OperationOutcome",
   "text": {"div": "The time slot you selected is already taken. Sorry for the inconvenience. Next availabilities :
DateTime-1 : 2023-03-20 13:00:00 DateTime-2 : 2023-03-20 13:15:00 DateTime-3 : 2023-03-20 13:30:00"}
}
```

Cancelling appointments

HTTP method:
PATCH

URL:
v1 = https://{FQDN}/apps/api/v1/fhir/{PracticeCode}/dstu2/Appointment/{AppointmentIdentifier}
v2 = https://{FQDN}/apps/api/v1/fhir/{PracticeCode}/dstu2/v2/Appointment/{AppointmentIdentifier}

Headers:

Authorization: <OAuth2 Bearer Token>
Content-Type: application/fhir+json

Request Body:

```
[{
  "op": "replace",
  "path": "/status",
  "value": "cancelled"
}]
```

Sample:

Request Headers:

Accept: application/json+fhir
Authorization: <OAuth2 Bearer Token>

PATCH http://healow-preprod.healow.com/apps/api/v1/fhir/IFDECD/fhir/Appointment/889f6f9b-05cc-4bb8-8206-67eb07faaad9

Body:

```
[{
  "op": "replace",
  "path": "/status",
  "value": "cancelled"
}]
```

Error Handling:

In the event of an error, the system will respond by supporting below error codes.

HTTP Code	Severity	Description	Example
500	error	Forbidden	Invalid Access Token
401	error	Unauthorized	Access token not present.
400	error	OperationOutcome	EMR doesn't support functionality.
40	error	OperationOutcome	Invalid Appointment id

500	error	OperationOutcome	The appointment does not exist (deleted) or is not in the 'pending' s tatus.
400	error	Bad request	The appointment is either locked or a payment is a ssociated with it.

Sample error response:


```
{"error": {
   "message": "An access token is required to request this resource.",
   "type": "OAuthException",
   "reason": "unauthorized"
}}
```

Retrieve Tele-visit link

HTTP method:
GET

URL:
https://{FQDN}/apps/api/v1/fhir/{PracticeCode}/dstu2/Appointment?_id={AppointmentIdentifier}&type=VR

Request Headers:
Authorization: <OAuth2 Bearer Token>
Content-Type: application/fhir+json

Request Parameters:

Name	Required?	Type	Description
_id	yes	String	Appointment identifier
type	yes	String	Type of appointment (should be defaulted to "VR" for tele visit appointment type)

FHIR Resource Schema

Element Name	Data Type	Card.	Description & Constraints
id	string	1..1	Appointment id
telehealth-videolink extension	object	1..1	Tele visit link URL

Sample:

Request Headers:
Accept: application/json+fhir
Authorization: <OAuth2 Bearer Token>

GET http://healow-preprod.healow.com/apps/api/v1/fhir/IFDECD/fhir/Appointment?_id=889f6f9b-05cc-4bb8-8206-67eb07faaad9&type=VR

Response:

```
{
  "resourceType": "Bundle",
  "id": "f3d5ece4-5fbf-4ab5-af75-cbd7238b2b07",
  "meta": {"lastUpdated": "2023-11-28T10:38:21.958-05:00"},
  "type": "searchset",
  "total": 1,
  "link": [{
    "relation": "self",
    "url": " http://healow-preprod.healow.com/apps/api/v1/fhir/IFDECD/fhir/Appointment?_id=kfqHBSUlGsAfrOeAPFALUwKDiPeYaU-H4LC7qZXAoI4&practice_code=ACEBCD&type=VR"
  }],
  "entry": [{
    "fullUrl": " http://healow-preprod.healow.com/apps/api/v1/fhir/IFDECD/fhir/Appointment/kfqHBSUlGsAfrOeAPFALUwKDiPeYaU-H4LC7qZXAoI4",
```

```
      "resource": {
        "resourceType": "Appointment",
        "id": "kfqHBSUlGsAfrOeAPFALUwKDiPeYaU-H4LC7qZXAoI4",
        "extension": [{
          "url": "http://hl7.org.au/fhir/StructureDefinition/telehealth-videolink",
          "valueUri": " http://healow-preprod.healow.com/apps/tv/custom/1WY-0LH-PZQ-QPOD"
        }],
        "type": {"coding": [{
          "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
          "code": "VR",
          "display": "virtual"
        }]}
      }
    }]
}
```


Error Handling:

In the event of an error, the system will respond by supporting below error codes.

HTTP Code	Severity	Description	Example
500	error	Forbidden	Invalid Access Token
401	error	Unauthorized	Access token not present.
400	error	OperationOutcome	EMR doesn't support functionality.
400	error	OperationOutcome	Invalid Appointment id
500	error	OperationOutcome	The appointment does not exist (deleted) or is not in the 'pending' s tatus.
400	error	Bad request	The appointment is locked.
400	error	Bad request	The tele visit link is not available.

Sample error response:


```
{"error": {
  "message": "An access token is required to request this resource.",
  "type": "OAuthException",
  "reason": "unauthorized"
}}
```